## Sections 4.3 and 4.4 - The Chain Rule

**The Chain Rule:** If $g$ is differentiable at $x$ and $f$ is differentiable at $g(x)$, then the composite function $m(x) = f(g(x))$is differentiable at $x$ and is given by

$$\boxed{m'(x) = f'(g(x)) \cdot g'(x)}$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}$$

$\log(x^7)$
$= 7\log(x)$

$\log(2x+1)$

**General Derivative Rules:**

- If $y = [f(x)]^n$ then $y' = n[f(x)]^{n-1} \cdot f'(x)$ $\qquad y = x^n \quad y' = nx^{n-1}$
- If $y = e^{f(x)}$ then $y' = e^{f(x)} \cdot f'(x)$ $\qquad y = e^x \quad y' = e^x$
- If $y = b^{f(x)}$ then $y' = \ln b \cdot b^{f(x)} \cdot f'(x)$ $\qquad y = b^x \quad y' = \ln b \cdot b^x$
- If $y = \ln(f(x))$ then $y' = \frac{1}{f(x)} \cdot f'(x)$ $\qquad y = \ln(x) \quad y' = \frac{1}{x}$
- If $y = \log_b(f(x))$ then $y' = \frac{1}{\ln b} \cdot \frac{1}{f(x)} \cdot f'(x)$ $\qquad y = \log_b(x) \quad y' = \frac{1}{\ln b} \cdot \frac{1}{x}$

**Example 1:** Differentiate the following:

a) $f(x) = (4x^2 + 7x)^5$

$$\boxed{f'(x) = 5(4x^2+7x)^4 \cdot (8x+7)}$$

b) $g(x) = 6(x^{1/2} - 3x)^4$

$$\boxed{g'(x) = 6 \cdot 4(x^{1/2} - 3x)^3 \cdot \left(\tfrac{1}{2}x^{-1/2} - 3\right)}$$

c) $y = \dfrac{5}{(t^2 + 3t + 4)^4} = 5(t^2+3t+4)^{-4}$

$$\boxed{y' = 5 \cdot -4(t^2+3t+4)^{-5} \cdot (2t + 3 + 0)}$$

d) $F(x) = e^{x^2}$

$$\boxed{F'(x) = e^{x^2} \cdot 2x}$$

e) $H(x) = 3x^5 e^{x^4}$

$$\boxed{H'(x) = (3x^5)\left(e^{x^4} \cdot 4x^3\right) + e^{x^4}\left(15x^4\right)}$$

f) $h(x) = \dfrac{3}{\sqrt[4]{(3x+2)^5}} = \dfrac{3}{(3x+2)^{5/4}} = 3(3x+2)^{-5/4}$

$$\boxed{h'(x) = 3 \cdot \tfrac{-5}{4}(3x+2)^{-9/4} \cdot 3}$$

g) $f(x) = (4x^2+5)^6\left(\sqrt[3]{(3x^4-5x+7)^4}\right) = (4x^2+5)^6(3x^4-5x+7)^{4/3}$

$$\boxed{f'(x) = (4x^2+5)^6 \cdot \tfrac{4}{3}(3x^4-5x+7)^{1/3} \cdot (12x^3-5) + (3x^4-5x+7)^{4/3} \cdot 6(4x^2+5)^5 \cdot 8x}$$

$\log(AB) = \log(A) + \log(B)$

$\log\left(\frac{A}{B}\right) = \log(A) - \log(B)$

$\log(A^B) = B\log(A)$

h) $y = \log_8\left(\sqrt{1+x^2} + 10x\right) = \log_8\left((1+x^2)^{1/2} + 10x\right)$

$$\boxed{y' = \frac{1}{\ln 8} \cdot \frac{1}{(1+x^2)^{1/2} + 10x} \cdot \left(\tfrac{1}{2}(1+x^2)^{-1/2} \cdot 2x + 10\right)}$$

i) $y = \log_7\left(\dfrac{x-1}{x+2}\right) = \log_7(x-1) - \log_7(x+2)$

$$\boxed{y' = \frac{1}{\ln 7} \cdot \frac{1}{x-1} \cdot 1 - \frac{1}{\ln 7} \cdot \frac{1}{x+2} \cdot 1}$$

j) $y = \dfrac{5\ln((x^2+x)^5)}{x^3} = \dfrac{25\ln(x^2+x)}{x^3} = 25\ln(x^2+x) \cdot x^{-3} = 25x^{-3}\ln(x^2+x)$

OR// $25\ln(x^2+x) \cdot x^{-3} = 25x^{-3}\ln(x^2+x)$

$$\boxed{y' = \frac{x^3 \cdot \left(25 \cdot \frac{1}{x^2+x} \cdot (2x+1)\right) - \left(25\ln(x^2+x)\right) \cdot 3x^2}{(x^3)^2}}$$

**Example 2:** Find the value(s) of $x$ where the tangent line is horizontal for $f(x) = \frac{x^2}{(2-3x)^3}$

$(f'(x)=0)$

$$f'(x) = \frac{(2-3x)^3 \cdot 2x - x^2 \cdot 3(2-3x)^2 \cdot -3}{((2-3x)^3)^2}$$

$$= \frac{2x(2-3x)^3 + 9x^2(2-3x)^2}{(2-3x)^6}$$

$$= \frac{x\cancel{(2-3x)^2}(2(2-3x)+9x)}{(2-3x)^{\cancel{6}\,4}}$$

$$= \frac{x(4-6x+9x)}{(2-3x)^4} = \frac{x(4+3x)}{(2-3x)^4}$$

$f'(x)=0$

$$\frac{x(4+3x)}{(2-3x)^4} = 0$$

$x(4+3x)=0$

$\boxed{x=0}$ $\quad 4+3x=0$

$3x=-4$

$\boxed{x=-4/3}$

*recall a fraction is zero when its numerator is zero*

**Example 3:** You are given the following information: $g(4) = 8$, $g'(4) = 3$, $f'(4) = 7$, $f'(8) = 6$, $f'(3) = 5$. If $h(x) = f(g(x))$, what is $h'(4)$?

$h(x) = f(g(x))$

$h'(x) = f'(g(x)) \cdot g'(x)$

$h'(4) = f'(g(4)) \cdot g'(4)$

$= f'(8) \cdot 3 = 6 \cdot 3 = \boxed{18}$

$f(x) = (4x+2)^6$

**Example 4:** Let $y = \ln u$ and $u = 5x^4 + x^6$. Find $\frac{dy}{dx}$ using Form 2 of the chain rule. Then, use Form 1 to compute the derivative and compare your answers.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$= \frac{1}{u} \cdot (20x^3 + 6x^5)$$

$$= \boxed{\frac{1}{5x^4+x^6} \cdot (20x^3+6x^5)}$$

$y = \ln(5x^4+x^6)$

$$\boxed{y' = \frac{dy}{dx} = \frac{1}{5x^4+x^6} \cdot (20x^3+6x^5)}$$

**Example 5:** Keith invests \$5,000 into a savings account offering interest at an annual rate of 2.4% compounded continuously. How fast is the balance growing after 8 years?

$P = 5000$

$r = .024$

$A(t) = Pe^{rt}$

$A(t) = 5000e^{.024t}$

$A'(t) = 5000e^{.024t} \cdot .024$

$A'(t) = 120e^{.024t}$

$A'(8) = 120e^{.024(8)} \approx \boxed{\$145.40/\text{year}}$

**Highly Suggested Homework Problems - Section 4.3: 1-37(odd), 41, 45 and Section 4.4: 1-73(odd)**